REPORT

Day 1 — From the outside to the product architecture

1. **Physical Inspection — Studying the Product**
2. Start with the device **without opening it**.

Students should document:

- Product name/model - Nokia 105 TA-1447
- code 11SIPS4INOO
- plastic body
- 73 GMS
- 119.00x49.20x14.40 mm
- developed by hmd global
- Manufacturer - Foxconn
- Approximate age/version - First released September 2019

  Predecessor Nokia 105 (2017)

  Successor Nokia 105 (2022)
- External dimensions - 119.0×49.2×14.4 mm (4.69×1.94×0.57 in)
- Weight 73.02 g (2.576 oz)
- Operating system - Series 30+
- Buttons/switches - the metal dome sheet in the first photo. These are mechanical switches that act as input sensors
- LEDs/displays – 1(one)
- Connectors - 3
- Power source – Lithium battery
- Memory 4MiB
- Storage 4MiB
- Battery 800 mAh, Li-ion
- Sensors - **Battery temperature sensing** : the 3-pin battery connector usually has a third pin wired to a thermistor in the battery pack. The sensor is in the battery, not on this board , **Microphone** : the small round capsule on red/black wires, soldered to the MIC pads. This is the only dedicated sensor.
- Mainboard: SIM slot, microSD slot, 3-pin battery connector, micro-USB port, 3.5 mm audio jack, and a mic wired to the MIC pads.

- Display: a small TFT module (marked TD-TNQ1805-1) connects through a 16-pin flex connector. Its backlight is a separate LED on twisted red/black wires, soldered to the pads marked LED (- / +).
- Speaker: soldered to the +SPK- pads.
- Keypad side: the metal dome sheet is visible in the first photo.

Then ask three questions:

A. What is this product supposed to do?

Example: A **Nokia phone** → provide reliable calling and messaging, with basic features such as contacts, alarm, FM radio, and long battery life.

B. What does the user expect?

For a Nokia phone:
Portable → durable → easy to use → long battery life → clear calling → reliable keypad → good network connectivity → loud speaker → simple interface

C. What engineering decisions can we already infer?

For example:

Keypad → input circuit → processor → SIM/network module → audio circuit → display → Li-ion battery → charging circuit → antenna

You haven't opened the product yet, but you are already forming a **functional hypothesis**. That is the essence of **reverse engineering**.

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2. Product Literature Review

Now move from **physical observation → historical research**.

Don't just search for the current product.

Look for:

Version 1 → Version 2 → Version 3 → current version

For each version, investigate:

Parameter	Older Nokia 105	Nokia 105 Plus TA-1447
Processor	Basic feature-phone processor	MediaTek MT6261D
Battery	Lower-capacity removable battery	800 mAh removable Li-ion battery
Display	Small monochrome/basic LCD	1.77-inch QQVGA colour display
Connectivity	Basic 2G cellular	2G GSM + Micro-USB 1.1 + 3.5 mm audio jack
Charging	older charging connector	Micro-USB charging

Parameter	Older Nokia 105	Nokia 105 Plus TA-1447
Sensors	Very limited / no major sensors	No major dedicated sensors
PCB technology	Larger,simpler feature-phone PCB	More compact integrated PCB with processor, power, audio and RF circuit
Power consumption	Very low	Low power,designed for long battery life
Safety features	Keypad lock, emergency calling, security code, battery/charger safety, restricted-area warnings, road-safety guidance	Keypad lock, emergency calling, security features, battery/charger safety, restricted-area warnings, road-safety guidance

Then ask:

Why did the manufacturer change it?

That question takes students beyond component identification into **engineering reasoning**.

For the **Nokia 105**, we can compare an older version with the newer **Nokia 105 TA-1447**.

For example:

Old: Basic monochrome display
↓
New: Colour display

Possible reasons:

- better **visual experience**
- easier to read menus and information
- supports additional features such as **games and multimedia**
- improves overall **user experience**
- meets changing **customer expectations**

Old: Smaller/older battery design
↓
New: 800 mAh removable battery

Possible reasons:

- longer operating time
- supports additional features
- maintains the phone's **low-power design**
- provides convenient battery replacement

Old: Older charging connector
↓
New: Micro-USB

Possible reasons:

- easier charging
- more widely available charging cables
- improved compatibility with accessories
- simpler connection

Important point:

Always distinguish **evidence** from **hypothesis**.

Evidence: What we can directly confirm from specifications, manuals, or physical observation.
Hypothesis: Our engineering explanation for **why** the manufacturer made that change.

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3. Product Analysis — Group Discussion

Now divide students into groups.

Give them three questions:

Purpose

What problem is this product solving?

- Provides **basic and reliable communication**
- Allows users to make calls and send messages
- Provides essential features without unnecessary complexity
- Offers communication with **low battery consumption**
- Provides an affordable phone option

User expectations

What would users consider a good product?

user would expect:

- **Long battery life**
- **Clear call quality**
- **Strong network connectivity**
- **Easy-to-use keypad**
- **Durable body**
- **Clear and readable display**
- **Loud speaker**

- **Affordable price**
- **Reliable performance**
- **Simple interface**

Trade-offs

What compromises did the designer probably make?

o keep the phone **affordable, simple and power-efficient**, the designers probably accepted:

- Basic processor → **lower performance but lower power consumption**
- Small display → **less multimedia capability but longer battery life**
- Basic camera/features → **lower cost**
- Plastic body → **lighter and cheaper, but less premium**
- 2G connectivity → **lower complexity and power consumption but slower data**
- Limited storage → **lower cost and simpler hardware**

Main idea: The Nokia 105 is designed by balancing **cost, battery life, durability, simplicity and functionality** rather than maximizing every feature.

But engineering creates conflicts:

Higher performance → more power → more heat → larger cooling requirement.

Larger battery → longer runtime → increased weight/thickness.

So reverse engineering isn't merely:

"What components are inside?"

It is important to move beyond simply identifying the components and understand:

"Why was the product designed this way?"

This helps us understand the **engineering decisions** behind the product, such as:

- Why is the battery removable?
- Why is the display small?
- Why is the keypad physical?
- Why are only basic features included?
- Why is the PCB compact?
- Why is the phone designed for low power consumption?

Understanding these decisions is a much more valuable **engineering skill** than simply knowing the names of the components.

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4. Introduction to Guided Teardown

Only after completing **documentation and research** should students open the Nokia 105.

The sequence should be:

Document → Photograph → Open → Photograph → Identify → Record

Important:

Never immediately start removing components.

First observe and document the phone's construction. Then carefully open it and record each component and its position.

Tools

Typical teardown kit:

- **Precision screwdriver set** → to remove small screws
- **Plastic pry tools** → to separate the phone casing safely
- **Spudger** → to lift connectors and small components
- **Tweezers** → to handle small parts
- **ESD wrist strap** → to protect components from static electricity
- **ESD mat** → to provide a static-safe working surface

For Nokia 105 TA-1447, focus on identifying:

- Battery
- Display
- Keypad/contact board
- Main PCB
- Speaker
- Microphone
- Charging port
- SIM card section
- Antenna/RF section
- Buttons and connectors
- Screws and casing structure

Goal: Don't just take the phone apart—understand **how each component contributes to the overall function of the Nokia 105.**

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5. ESD Precautions

tudents should understand why **ESD (Electrostatic Discharge)** matters during the teardown of the **Nokia 105 TA-1447**.

A person can carry a static electrical charge without noticing it. This charge can damage sensitive electronic components on the phone's PCB.

Simple rule:

Before touching the PCB:

Ground yourself → Use ESD protection → Handle the PCB by its edges → Avoid touching IC pins and contacts unnecessarily

Power Safety

Before opening the Nokia 105:

- **Switch off the phone completely.**
- **Remove the battery before handling the PCB.**
- Do not short-circuit the battery terminals.
- Handle the battery carefully and avoid bending, puncturing, or damaging it.
- Do not work on the phone while it is connected to a charger.

The main purpose is to prevent **ESD damage, short circuits, and battery-related hazards**.

Important: A Nokia 105 uses a low-voltage battery, but the battery can still deliver enough current to cause damage or heating if its terminals are shorted.

Day 2 — Guided Teardown

Now the real reverse engineering begins.

Step 1 — Opening the device

The instructor should demonstrate first.

Students record:

- Screw locations
- Connector locations
- Cable routing
- Mechanical clips
- PCB mounting points
- Shields
- Battery location
- Antenna location
- Heat sinks
- Grounding points

Very important:

Photograph before removing anything.

Then photograph again after each major layer is removed.

This creates a **disassembly record**.

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6. Internal Hardware Analysis — Part I

Now create a component inventory.

Don't ask students merely:

"What is this component?"

Instead ask them to build a structured table.

Ref./Location	Component	Package/Type	Marking	Function	Notes
U1	Main processor / SoC	—	As observed	Controls phone operations	Main control IC
U2	Power-management IC	—	As observed	Regulates power to different circuits	Connected to battery/power section
C1	Capacitor	SMD	—	Filtering / decoupling	Located near ICs
R1	Resistor	SMD	—	Current limiting / signal control	Value depends on marking
J1	Charging connector	Micro-USB	—	Battery charging / connection	External connector

This transforms the teardown into **engineering data collection**.

Day 3 — Component Markings, Ratings & Specifications

This is where students move from **identification to datasheet-based reverse engineering**.

Suppose they find an IC marked:

STM32F103C8T6

They should determine:

- Manufacturer
- Part number
- MCU family
- CPU architecture
- Flash
- RAM
- Operating voltage
- Package

- GPIO count
- Communication peripherals
- ADC
- Maximum clock frequency

Then connect the datasheet information back to the physical PCB.

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7. Component Functional Analysis

This is probably the **most important activity** in the whole module.

Don't stop at:

"U1 is an STM32."

Ask:

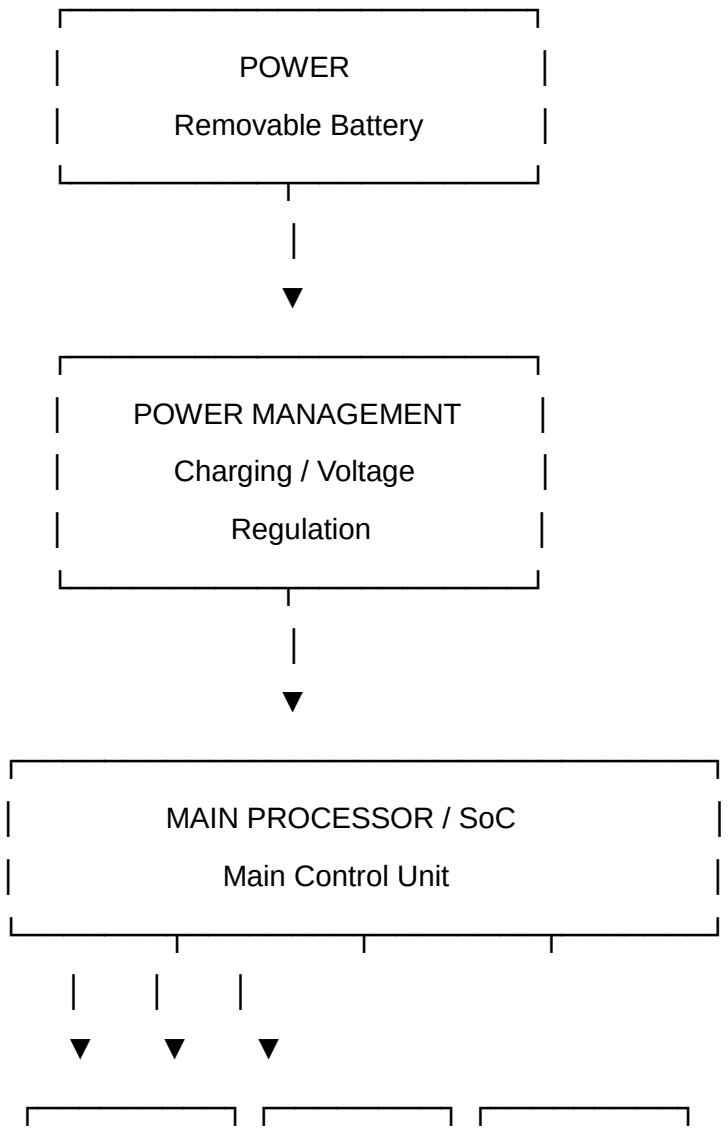
"What role does U1 play in this product?"

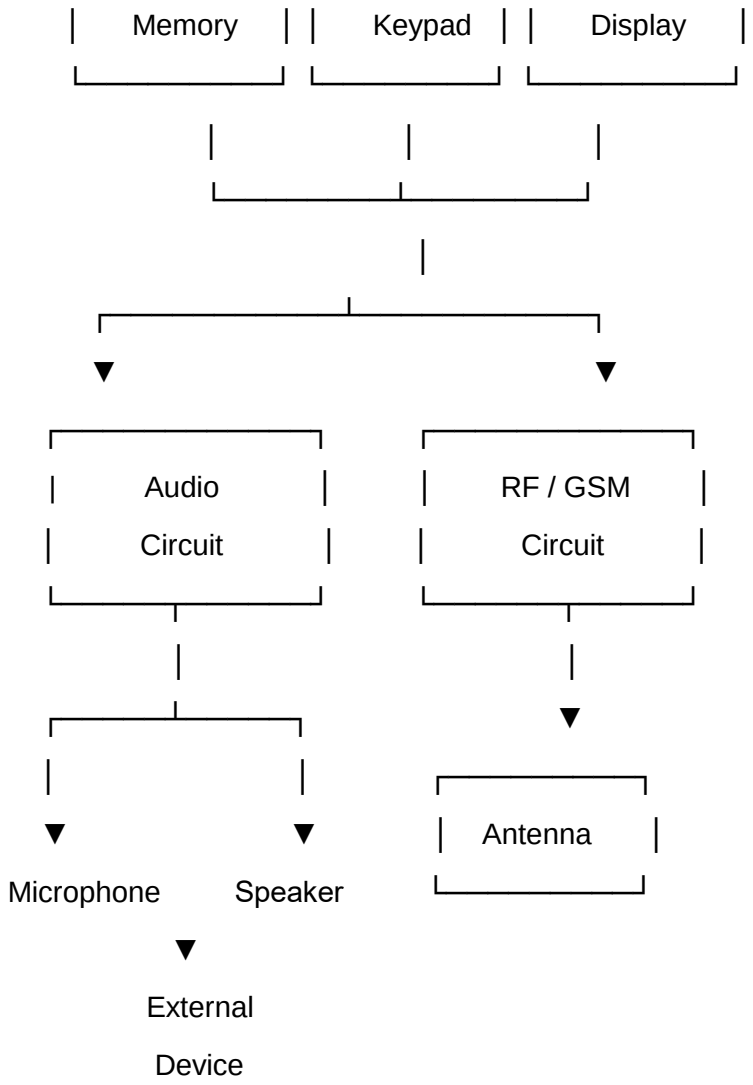
Create a **Function Table**.

Component	Electrical Role	Functional Role in Nokia 105	Connected To
Main Processor / SoC	Digital processing	Controls phone operations, calls, display and keypad	Display, keypad, memory, RF circuits
Power Management IC	Voltage regulation	Manages and distributes power to different circuits	Battery → phone circuits
MOSFETs	Electronic switching	Controls power paths and charging circuits	Power-management section
Crystal / Oscillator	Clock source	Provides timing reference for processor and communication circuits	Main processor
Memory	Data storage	Stores software, settings and user data	Main processor
Display	Visual output	Shows menus, numbers, messages and status	Main processor
Keypad	Input signal	Allows the user to enter numbers and control the phone	Main processor
Microphone	Audio input	Converts the user's voice into an electrical signal	Audio circuit / processor

Component	Electrical Role	Functional Role in Nokia 105	Connected To
Speaker	Audio output	Produces voice, ringtone and alert sounds	Audio circuit
RF/Network Circuit	Radio-frequency processing	Enables communication with the cellular network	Antenna ↔ processor
Antenna	RF transmission/reception	Sends and receives wireless signals	RF circuit
Capacitors	Filtering / decoupling	Reduce noise and stabilize power and signals	ICs and power lines
Resistors	Current/signal control	Set voltages and control signal levels	Various circuits

Example architecture





Now the students are no longer looking at a PCB as a collection of components.

They are seeing it as a **system architecture**.

The key skill I would emphasize

Your **Nokia 105 reverse-engineering module** should progressively develop these five levels of thinking:

Level 1 — Observation

"I can see a removable battery, keypad, display, PCB and several ICs."

↓

Level 2 — Identification

"This component is the main processor, and this section is the power-management circuit."

↓

Level 3 — Specification

“The component's datasheet tells us its operating voltage, package, electrical ratings and supported functions.”

↓

Level 4 — Functional Understanding

“The main processor controls the keypad, display, memory, audio and communication functions.”

↓

Level 5 — Engineering Reasoning

“The manufacturer selected these components and this architecture to achieve **low cost, long battery life, durability, simple operation and reliable communication**.”

The final goal

Observe → Identify → Find specifications → Understand function → Explain engineering decisions

This is the core skill developed through **reverse engineering**.

Recommended student deliverables

By the end of these three days, every team should produce:

1. **External Product Inspection Sheet**
2. **Product Version/Literature Comparison**
3. **User Needs & Trade-off Analysis**
4. **Teardown Photo Log**
5. **Mechanical Disassembly Sequence**
6. **Component Inventory/BOM**
7. **Component Marking & Datasheet Table**
8. **Functional Block Diagram**
9. **Component Function Table**
10. **Final Reverse-Engineering Report**

And I would add one more deliverable:

"Reverse Engineering Conclusions"

Students should answer:

What did we learn about the designer's decisions that was not obvious from simply looking at the product?

From studying and analyzing the **Nokia 105 TA-1447**, we learned that:

- The phone was designed for **low power consumption and long battery life**.
- A **simple processor and limited features** help keep the phone affordable and power-efficient.
- The **physical keypad** was chosen for simple and reliable operation.
- The small display reduces **power consumption** while providing the information users need.
- The internal PCB is designed to **fit many functions into a compact space**.
- The power-management circuit carefully distributes power to different sections of the phone.
- The RF and antenna design allows the phone to provide **reliable cellular communication**.
- The removable battery makes the phone **easier to maintain and replace**.
- The overall design shows a deliberate balance between **cost, durability, simplicity, battery life and functionality**.

Final conclusion:

“By looking inside the Nokia 105, we learned that its simple appearance is the result of many engineering decisions made to achieve reliability, low cost, low power consumption and ease of use.”

Images of phone are shown below:





